

1.1 Image formats

Let's start with the very basics—image formats. There are several image formats available for storing all those great photos you take. This section examines them in brief.

1.1.1. BMP

BMP or bitmap is a raster format, which is used practically for all possible raster data storage. It is device independent, that is, it determines the pixel's colour without reference to the display device and enables you to record images with varying quality levels (to the point of 32 bits).

The main advantage of this format is its usability and wide software support. The compression algorithm used in BMP, can be one of the RLE run-length encoding variants. However, most files do not use compression.

1.1.2. JPEG

This format was designed to transfer graphic data and images over digital networks and was generally used to hold and transfer full colour photorealistic images. Before JPEG, there were very few formats, which supported 24-bit halftone images. TIFF and BMP formats allowed holding 24-bit data, but they failed to perform lossless compression of the data, which contained 1000 colours from the real world, on the high quality level.

JPEG compresses photos, though with quality loss. Photos and multi-colour images, transferred in this format, are ideal for networking. It is not possible to refine JPEG images, though it is possible to debase them by decreasing the file size. This format is primarily used to hold photorealistic graphic images with a large number of colours. Compressed data capacity depends on image contents.

JPEG supports Progressive JPEG standard, which conceptually looks like interlaced in the GIF format. This means that the JPEG